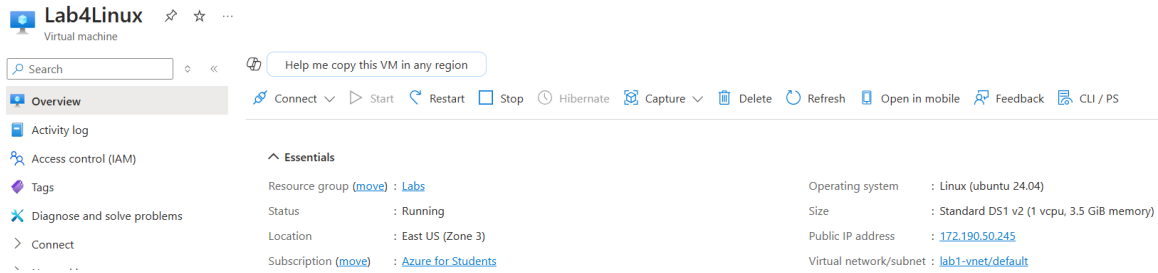


Juan Pablo de Legarreta
NETC-240
Lab 4: 2-5-2025

Document each step of the lab. Take the appropriate screenshots and annotate them.

1. Create a Ubuntu 24.04 LTS VM. Create a new user using your first name as the name of the user. Use a 2nd SSH session to login using your first name account.



Created a Ubuntu machine named Lab4Linux.

```
azureuser@Lab4Linux:~$ sudo adduser juan_pablo
info: Adding user `juan_pablo' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `juan_pablo' (1001) ...
info: Adding new user `juan_pablo' (1001) with group `juan_pablo (1001)' ...
info: Creating home directory `/home/juan_pablo' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for juan_pablo
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n]
info: Adding new user `juan_pablo' to supplemental / extra groups `users' ...
info: Adding user `juan_pablo' to group `users' ...
azureuser@Lab4Linux:~$
```

Added user juan_pablo.

```

PS C:\Users\JP> ssh juan_pablo@172.190.50.245
(juan_pablo@172.190.50.245) Password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-1021-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Feb  5 22:13:24 UTC 2025

juan_pablo@Lab4Linux:~$ |

```

Screenshots of logging in through ssh to the juan_pablo account and the CLI for the juan_pablo account.

2. Run the Lab4.sh script in your home directory

```

juan_pablo@Lab4Linux:~$ #!/bin/bash

# Create main USA directory
mkdir USA

# Create state directories
states=(Alabama Alaska Arizona Arkansas California Colorado Connecticut Delawa
re Florida Georgia Hawaii Idaho Illinois Indiana Iowa Kansas Kentucky Louisian
a Maine Maryland Massachusetts Michigan Minnesota Mississippi Missouri Montana
Nebraska Nevada NewHampshire NewJersey NewMexico NewYork NorthCarolina NorthD
akota Ohio Oklahoma Oregon Pennsylvania RhodeIsland SouthCarolina SouthDakota
Tennessee Texas Utah Vermont Virginia Washington WestVirginia Wisconsin Wyomin
g)

for state in ${states[@]}; do
    mkdir "USA/$state"
done

# Create top city files
touch USA/Alabama/{Birmingham, Montgomery, Mobile, Huntsville, Tuscaloosa}
touch USA/Alaska/{Anchorage, Juneau, Fairbanks, Wasilla, Ketchikan}
touch USA/Arizona/{Phoenix, Tucson, Mesa, Chandler, Gilbert}
touch USA/Arkansas/{LittleRock, FortSmith, Fayetteville, Springdale, Rogers}

```

A screenshot of running the Lab4 bash script, with the confirmation of the USA directory, below, shown with the ls -la command.

```

juan_pablo@Lab4Linux:~$ ls -la
total 32
drwxr-x--- 4 juan_pablo juan_pablo 4096 Feb  5 22:15 .
drwxr-xr-x 4 root      root      4096 Feb  5 22:07 ..
-rw----- 1 juan_pablo juan_pablo   66 Feb  5 22:11 .bash_history
-rw-r--r-- 1 juan_pablo juan_pablo  220 Feb  5 22:07 .bash_logout
-rw-r--r-- 1 juan_pablo juan_pablo 3771 Feb  5 22:07 .bashrc
drwx----- 2 juan_pablo juan_pablo 4096 Feb  5 22:13 .cache
-rw-r--r-- 1 juan_pablo juan_pablo  807 Feb  5 22:07 .profile
drwxrwxr-x 52 juan_pablo juan_pablo 4096 Feb  5 22:15 USA

```

3. Use the cd command to navigate into the "California" subdirectory.

```
juan_pablo@Lab4Linux:~$ cd USA/California
juan_pablo@Lab4Linux:~/USA/California$ |
```

Used cd USA/California to navigate to the California subdirectory

4. Use mkdir to create a subdirectory called "San Francisco"

```
juan_pablo@Lab4Linux:~/USA/California$ mkdir SanFrancisco
mkdir: cannot create directory 'SanFrancisco': File exists
juan_pablo@Lab4Linux:~/USA/California$ mkdir San_Francisco
juan_pablo@Lab4Linux:~/USA/California$ |
```

Tried to make a SanFrancisco directory, but it already existed so I made one called San_Francisco.

5. Use touch to create a file called "goldengate.txt" inside the San Francisco subdirectory.

```
juan_pablo@Lab4Linux:~/USA/California$ touch San_Francisco/goldengate.txt
juan_pablo@Lab4Linux:~/USA/California$ ls
Fresno LosAngeles SanDiego SanFrancisco SanJose San_Francisco
juan_pablo@Lab4Linux:~/USA/California$ cd San_Francisco
juan_pablo@Lab4Linux:~/USA/California/San_Francisco$ ls
goldengate.txt
juan_pablo@Lab4Linux:~/USA/California/San_Francisco$ |
```

Added the goldengate.txt file in the San_Francisco directory, shown with ls.

6. Remove the San Francisco subdirectory and goldengate.txt file you just created using the rmdir command. What switches are required to accomplish this with rmdir?

```
juan_pablo@Lab4Linux:~/USA/California$ rmdir San_Francisco
rmdir: failed to remove 'San_Francisco': Directory not empty
juan_pablo@Lab4Linux:~/USA/California$ rm -r San_Francisco
juan_pablo@Lab4Linux:~/USA/California$ ls
Fresno LosAngeles SanDiego SanFrancisco SanJose
juan_pablo@Lab4Linux:~/USA/California$ |
```

So the rmdir command was unable to delete the directory, as it was not empty. None of the tags associated with rmdir could accomplish this, and research showed that you can use the rm command instead, with the switch -r for recursive, which deletes anything within a directory along with the directory.

7. While inside the USA directory, list the contents using ls to see all the state subdirectories.

```

juan_pablo@Lab4Linux:~/USA/California$ cd ..
juan_pablo@Lab4Linux:~/USA$ ls
Alabama      Indiana      Nebraska     SouthCarolina
Alaska       Iowa         Nevada       SouthDakota
Arizona      Kansas       NewHampshire Tennessee
Arkansas     Kentucky    NewJersey    Texas
California   Louisiana   NewMexico    Utah
Colorado     Maine       NewYork      Vermont
Connecticut  Maryland    NorthCarolina Virginia
Delaware     Massachusetts NorthDakota  Washington
Florida      Michigan    Ohio         WestVirginia
Georgia      Minnesota   Oklahoma     Wisconsin
Hawaii       Mississippi Oregon        Wyoming
Idaho        Missouri    Pennsylvania
Illinois     Montana     RhodeIsland
juan_pablo@Lab4Linux:~/USA$ |

```

Used the `cd ..` command to go up a directory from the California directory and end up in the USA directory. Then used the `ls` command to display all of the directories.

8. Navigate into the Texas subdirectory and use `ls` again to see the 5 city files.

```

juan_pablo@Lab4Linux:~/USA$ cd Texas
juan_pablo@Lab4Linux:~/USA/Texas$ ls
Austin  Dallas  FortWorth  Houston  SanAntonio
juan_pablo@Lab4Linux:~/USA/Texas$ |

```

Used the `cd` command to get to the Texas subdirectory and the `ls` command to show the files.

9. Use `cd ..` twice to go back up to the USA directory.

```

juan_pablo@Lab4Linux:~/USA/Texas$ cd ..
juan_pablo@Lab4Linux:~/USA$ cd ..
juan_pablo@Lab4Linux:~$ |

```

Using `cd ..` twice actually takes you back up to the home directory.

10. Create a single Bash script which performs the above operations in steps 2-9. Delete the USA dir, then run the Lab4.sh script again to re-create it. Show your script and the results

```

juan_pablo@Lab4Linux:~$ #!/bin/bash

# Create main USA directory
mkdir USA

# Create state directories
states=(Alabama Alaska Arizona Arkansas California Colorado Connecticut Delaware
Florida Georgia Hawaii Idaho Illinois Indiana Iowa Kansas Kentucky Louisiana Main
e Maryland Massachusetts Michigan Minnesota Mississippi Missouri Montana Nebraska
Nevada NewHampshire NewJersey NewMexico NewYork NorthCarolina NorthDakota Ohio O
klahoma Oregon Pennsylvania RhodeIsland SouthCarolina SouthDakota Tennessee Texas
Utah Vermont Virginia Washington WestVirginia Wisconsin Wyoming)

for state in ${states[@]}; do
    mkdir "USA/$state"
done

# Create top city files
touch USA/Alabama/{Birmingham,Montgomery,Mobile,Huntsville,Tuscaloosa}
touch USA/Alaska/{Anchorage,Juneau,Fairbanks,Wasilla,Ketchikan}

```

The first part of the bash script, unchanged.

```

mkdir USA/California/San_Francisco
touch USA/California/San_Francisco/goldengate.txt
ls USA/California
ls USA/California/San_Francisco
rm -r USA/California/San_Francisco
ls USA/California
cd USA/Texas
ls
Fresno  LosAngeles  SanDiego  SanFrancisco  SanJose  San_Francisco
goldengate.txt
Fresno  LosAngeles  SanDiego  SanFrancisco  SanJose
Austin  Dallas  FortWorth  Houston  SanAntonio
juan_pablo@Lab4Linux:~/USA/Texas$ |

```

The steps I added to the script, following steps 2-9 from the lab.

The output follows the script, per the steps I added. The `ls USA/California` step is to show the successful creation of the directory `San_Francisco`. The `ls USA/California/San_Francisco` is to show the successful creation of the `goldengate.txt` file. Then the next `ls USA/California` is used to display the successful deletion of the `San_Francisco` directory. The final part of the script changes to the Texas directory and displays its contents.

11. Repeat #10, but instead of a Bash script, create a textfile that has the same instructions as your script in #10. Use redirection of Stdin to read the commands in from the file.

```
GNU nano 7.2 bashfile.txt *
#!/bin/bash

# Create main USA directory
mkdir USA

# Create state directories
states=(Alabama Alaska Arizona Arkansas California Colorado Connecticut Delaware)

for state in ${states[@]}; do
    mkdir "USA/$state"
done

# Create top city files
touch USA/Alabama/{Birmingham, Montgomery, Mobile, Huntsville, Tuscaloosa}
touch USA/Alaska/{Anchorage, Juneau, Fairbanks, Wasilla, Ketchikan}
touch USA/Arizona/{Phoenix, Tucson, Mesa, Chandler, Gilbert}
touch USA/Arkansas/{LittleRock, FortSmith, Fayetteville, Springdale, Rogers}
touch USA/California/{LosAngeles, SanDiego, SanJose, SanFrancisco, Fresno}
touch USA/Colorado/{Denver, ColoradoSprings, Aurora, FortCollins, Lakewood}
touch USA/Connecticut/{Bridgeport, NewHaven, Hartford, Stamford, Waterbury}
touch USA/Delaware/{Wilmington, Dover, Newark, Bear, Middletown}
touch USA/Florida/{Jacksonville, Miami, Tampa, Orlando, StPetersburg}

mkdir USA/California/San_Francisco
touch USA/California/San_Francisco/goldengate.txt
ls USA/California
ls USA/California/San_Francisco
rm -r USA/California/San_Francisco
ls USA/California
ls USA/Texas
|

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

The top and bottom of the text file called 'bashfile.txt' to use for this step.

```
juan_pablo@Lab4Linux:~$ bash < bashfile.txt
Fresno LosAngeles SanDiego SanFrancisco SanJose San_Francisco
goldengate.txt
Fresno LosAngeles SanDiego SanFrancisco SanJose
Austin Dallas FortWorth Houston SanAntonio
juan_pablo@Lab4Linux:~$ ls
USA bashfile.txt
juan_pablo@Lab4Linux:~$ |
```

Redirected the contents to the bash command, and got the same output as before, showing the San_Francisco directory, the file within it, the results of deleting the San_Francisco directory in California and the contents of the Texas Directory.

12. Demonstrate the use of the cd command to navigate the USA folder. Specifically, show cd .., use cd in both absolute and relative paths

```

juan_pablo@Lab4Linux:~$ cd USA
juan_pablo@Lab4Linux:~/USA$ ls
Alabama      Hawaii      Massachusetts  NewMexico    SouthDakota
Alaska       Idaho       Michigan        NewYork      Tennessee
Arizona      Illinois    Minnesota       NorthCarolina Texas
Arkansas     Indiana     Mississippi     NorthDakota  Utah
California   Iowa        Missouri        Ohio          Vermont
Colorado     Kansas     Montana         Oklahoma     Virginia
Connecticut  Kentucky   Nebraska        Oregon        Washington
Delaware     Louisiana  Nevada          Pennsylvania WestVirginia
Florida      Maine      NewHampshire    RhodeIsland  Wisconsin
Georgia      Maryland   NewJersey       SouthCarolina Wyoming
juan_pablo@Lab4Linux:~/USA$ cd Idaho
juan_pablo@Lab4Linux:~/USA/Idaho$ ls
Boise IdahoFalls Meridian Nampa Pocatello
juan_pablo@Lab4Linux:~/USA/Idaho$ cd ../../
juan_pablo@Lab4Linux:~$ cd USA/Idaho
juan_pablo@Lab4Linux:~/USA/Idaho$ cd ..
juan_pablo@Lab4Linux:~/USA$ cd ..
juan_pablo@Lab4Linux:~$ |

```

```

juan_pablo@Lab4Linux:/$ cd home/juan_pablo/USA/Idaho
juan_pablo@Lab4Linux:~/USA/Idaho$ cd ../../
juan_pablo@Lab4Linux:~$ |

```

Used the cd command to go one directory at a time, using relative paths, showing the contents of the directory along the way, moving to the USA directory then the Idaho directory.

Then I used the absolute path command to go to the home/juan_pablo/USA/Idaho directory and then back to the home directory.

13. Write the command that will ping a website once and save the results to a file named step14.

```

juan_pablo@Lab4Linux:~$ echo "ping -c 1 google.com" > step14
juan_pablo@Lab4Linux:~$ ls
USA bashfile.txt step14
juan_pablo@Lab4Linux:~$ cat step14
ping -c 1 google.com
juan_pablo@Lab4Linux:~$ |

```

The command ping -c 1 google.com was written into a file named step14

14. Create a file in nano that has 10 different 2 digit values, which are also not in any numerical order. Name that file step14.txt Show the contents of the file

```
GNU nano 7.2 step14.txt
26
12
76
39
42
81
57
19
63
91
```

The nano file step14.txt, shown above.

```
juan_pablo@Lab4Linux:~$ cat step14.txt
26
12
76
39
42
81
57
19
63
91

juan_pablo@Lab4Linux:~$ |
```

The contents of the file step14.txt are shown on the command line.

15. Use the Sort command to sort those values in descending order and save the results to a file named step14.txt. Show the contents of the file

```
juan_pablo@Lab4Linux:~$ sort -nr step14.txt > step14.txt
juan_pablo@Lab4Linux:~$ cat step14.txt
juan_pablo@Lab4Linux:~$ nano step14.txt
```

I tried to replace the contents into the file using the > operator, but it didn't work. I also tried to pipe the output into the cat command, like in the following screenshot:

```
juan_pablo@Lab4Linux:~$ sort -nr step14.txt | cat > step14.txt
juan_pablo@Lab4Linux:~$ cat step14.txt
juan_pablo@Lab4Linux:~$ |
```


That didn't work either.

```
juan_pablo@Lab4Linux:~$ sort -nr < step14.txt | echo > step14.txt
juan_pablo@Lab4Linux:~$ cat step14.txt

juan_pablo@Lab4Linux:~$ |
```

I then tried to use piping with a couple of commands, as seen above and below, without success.

```
juan_pablo@Lab4Linux:~$ sort -nr < step14.txt | > step14.txt
juan_pablo@Lab4Linux:~$ cat step14.txt
```

Finally, I just used a transfer file called transfer.txt to complete this task:

```
juan_pablo@Lab4Linux:~$ sort -nr < step14.txt > transfer.txt
juan_pablo@Lab4Linux:~$ cat transfer.txt > step14.txt
juan_pablo@Lab4Linux:~$ cat step14.txt
91
81
76
63
57
42
39
26
19
12
```

16. Count the number of lines in the file data.txt

I created a file called data.txt and it contains the following:

```

juan_pablo@Lab4Linux:~$ cat data.txt
Learning Linux can be a challenge for those who are used to using a GUI, like
Windows. In fact, many people (including myself) when they start out with
Linux feel rather lost and don't know how to look for things.

One of the biggest challenges I remember when I first started using Linux was
how to look for and find files. In Windows, I had always used the search
function of the File Explorer, but Linux CLI offers no such luxury.

I had to learn how to use the following commands in Linux:

find
grep
man
ls
cd
sudo
echo
... and others!

One of the other powerful parts of Linux that I did not understand at first
was piping. Piping in Linux can make really sophisticated commands that can
help not just in automation but also in many applications of cybersecurity,
such as forensics and log management.

Linux can do some pretty powerful stuff!

It is an impressive operating system. Good for Linux!
juan_pablo@Lab4Linux:~$ |

```

After writing the file, I got the number of lines with the `wc -l` command:

```

juan_pablo@Lab4Linux:~$ wc -l data.txt
27 data.txt

```

The document has 27 lines.

17. Print only the first 3 lines of the file `data.txt`

```

juan_pablo@Lab4Linux:~$ head -n 3 data.txt
Learning Linux can be a challenge for those who are used to using a GUI, like
Windows. In fact, many people (including myself) when they start out with
Linux feel rather lost and don't know how to look for things.
juan_pablo@Lab4Linux:~$ |

```

You can use `head` to print the first three lines of a document.

18. Print all the lines that contain the word "linux" from the file `data.txt`

```

juan_pablo@Lab4Linux:~$ grep Linux data.txt
Learning Linux can be a challenge for those who are used to using a GUI, like
Linux feel rather lost and don't know how to look for things.
One of the biggest challenges I remember when I first started using Linux was
function of the File Explorer, but Linux CLI offers no such luxury.
I had to learn how to use the following commands in Linux:
One of the other powerful parts of Linux that I did not understand at first
was piping. Piping in Linux can make really sophisticated commands that can
Linux can do some pretty powerful stuff!
It is an impressive operating system. Good for Linux!
juan_pablo@Lab4Linux:~$ |

```

You can use the grep command to print out all the lines that contain Linux, which in this case were 9 lines.

19. Sort the contents of the file data.txt and print the sorted output

```

juan_pablo@Lab4Linux:~$ sort data.txt

... and others!
I had to learn how to use the following commands in Linux:
It is an impressive operating system. Good for Linux!
Learning Linux can be a challenge for those who are used to using a GUI, like
Linux can do some pretty powerful stuff!
Linux feel rather lost and don't know how to look for things.
One of the biggest challenges I remember when I first started using Linux was
One of the other powerful parts of Linux that I did not understand at first
Windows. In fact, many people (including myself) when they start out with
cd
echo
find
function of the File Explorer, but Linux CLI offers no such luxury.
grep
help not just in automation but also in many applications of cybersecurity,
how to look for and find files. In Windows, I had always used the search
ls
man
such as forensics and log management.
sudo
was piping. Piping in Linux can make really sophisticated commands that can
juan_pablo@Lab4Linux:~$ |

```

Using the sort command automatically displays the sorted text. It's changed everything to be alphabetical. An interesting consequence of this is that it put all of the empty lines up top – making a big gap between the command and the output!

20. Create the correct Grep command that will generate the results for each case below. Use the U.S. Constitution textfile. Show the results of each command.

- a. Find how many times "United States" occurs (case insensitive)

```
grep -c -i "united states" US_Constitution.txt
```

```
juan_pablo@Lab4Linux:~$ grep -c -i "united states" US_Constitution.txt
279
juan_pablo@Lab4Linux:~$ |
```

You can use the grep command with the switches -c and -i (-c for count, or how many occurrences the word has, -i allows for case insensitive searching)

- b. Print all sentences with the word "taxes" (ending in ".")

```
grep -i "taxes.*\.$" US_Constitution.txt
```

```
juan_pablo@Lab4Linux:~$ grep -i "taxes.*\.$" US_Constitution.txt
and collect taxes, duties, imposts, and..
power to lay and collect taxes, duties...
States. All taxes, duties.....
Income taxes. Congress shall have power to      16      .....
juan_pablo@Lab4Linux:~$ |
```

This one is interesting – you have to use -i to keep it case insensitive, as before, but then you have to add some regex syntax after the word. The * indicates that there can be any characters between taxes and the period. The period has a \ before it because the period usually has a function in regex syntax. Finally, the \$ indicates that the period must be at the end of a sentence.

- c. Retrieve sections talking about presidents, using either "president" or "executive"

```
grep -Ei "president|executive" US_Constitution.txt
```

```
juan_pablo@Lab4Linux:~$ grep -Ei "president|executive" US_Constitution.txt
executive, and judicial branches was a boldly original attempt
Washington, of Virginia, was unanimously elected President, and the
and Messrs. Mason and Randolph, of Virginia. The president of the
The President informed Congress, on the 28th of January, 1790, that
State, the Executive Authority thereof shall issue Writs of
Legislature of any State, the Executive thereof may make
\4\The Vice President of the United States shall be
President of the Senate, but shall have no Vote, unless they be
President pro tempore, in the Absence of the Vice President, or
when he shall exercise the Office of President of the United
Oath or Affirmation. When the President of the United States is
be presented to the President of the United States; If he
President within ten Days (Sundays excepted) after it shall
presented to the President of the United States; and before the
```

So the command would have the switches -i again for case insensitive, and -E, which allows for extended regex. Using this switch allows for the or operator | to be used, so it will look for president OR executive (president|executive)

d. Get lines about popular vote or electors in Article 2 Section 1

```
awk '/Article II/,/Section 2/ {if (/popular vote|electors/i) print}' US_Constitution.txt |  
grep -Ei "popular vote|electors"
```

```
juan_pablo@Lab4Linux:~$ awk '/Article II/,/Section 2/ {if (/popular vote|electors  
/i) print}' US_Constitution.txt | grep -Ei "popular vote|electors"  
Legislature thereof may direct, a Number of Electors, equal to  
\3\The Electors shall meet in their respective States, and  
Majority of the whole Number of Electors appointed; and if  
Votes of the Electors shall be the Vice President. But if there  
Electors, and the Day on which they shall give their Votes;  
juan_pablo@Lab4Linux:~$ |
```

This one took some research. Basically you use awk to search for specific text between the sections 'Article II' and 'Section 2', which follows Section 1. Then you use that text and grep "popular vote OR electors" using the switches -Ei to allow for case insensitive and the OR operator.

e. Find all lines containing "shall" followed by a verb starting with "appo" but not "appoint".

```
sed -n '/shall[[:space:]]\+appo[^[:space:]]*/{ /appoint/!p }' US_Constitution.txt
```

```
juan_pablo@Lab4Linux:~$ sed -n '/shall[[:space:]]\+appo[^[:space:]]*/{ /appoint/!  
p }' US_Constitution.txt  
juan_pablo@Lab4Linux:~$ |
```

This one also took some research. Basically, you use sed to search for the pattern "shall" followed by a space, followed by "appo", not followed by a space, where the word is not appoint, in the file. The -n switch means that the command will print only the relevant lines. It did not print anything, and a quick ctrl+f on the text file found that the "shall appo" pattern ONLY was found with "shall appoint", therefore there correctly is no output.

21. Pick a domain name that you are confident no one else in class will use, then use the Dig command to find the A, MX and any TXT records. Then demo the use of the nslookup command for a public IP address

The output from using the switch for A records:

```
juan_pablo@Lab4Linux:~$ dig tryhackme.com -t A

; <<>> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<>> tryhackme.com -t A
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 2801
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;tryhackme.com.                IN      A

;; ANSWER SECTION:
tryhackme.com.                225     IN      A      104.22.54.228
tryhackme.com.                225     IN      A      172.67.27.10
tryhackme.com.                225     IN      A      104.22.55.228

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Feb 06 02:44:19 UTC 2025
;; MSG SIZE rcvd: 90

juan_pablo@Lab4Linux:~$ |
```

Looks like the IP addresses are 104.22.54.228, 104.22.55.228, and 172.67.27.10.

The output from using the switch for MX records:

```
juan_pablo@Lab4Linux:~$ dig tryhackme.com -t MX

; <<>> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<>> tryhackme.com -t MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 31055
;; flags: qr rd ra; QUERY: 1, ANSWER: 5, AUTHORITY: 0, ADDITIONAL: 3

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;tryhackme.com.                IN      MX

;; ANSWER SECTION:
tryhackme.com.                78      IN      MX      10 alt3.aspmx.l.google.com.
tryhackme.com.                78      IN      MX      1 aspmx.l.google.com.
tryhackme.com.                78      IN      MX      10 alt4.aspmx.l.google.com.
tryhackme.com.                78      IN      MX      5 alt1.aspmx.l.google.com.
tryhackme.com.                78      IN      MX      5 alt2.aspmx.l.google.com.

;; ADDITIONAL SECTION:
aspmx.l.google.com.          78      IN      A      172.253.62.27
alt3.aspmx.l.google.com.    78      IN      A      142.250.27.26

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Feb 06 02:45:29 UTC 2025
;; MSG SIZE rcvd: 189
```

Looks like google hosts their email services based on the output.

The output from using the switch for TXT records:

```
juan_pablo@Lab4Linux:~$ dig tryhackme.com -t TXT

; <<>> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<>> tryhackme.com -t TXT
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 56927
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;tryhackme.com.                IN      TXT

;; ANSWER SECTION:
tryhackme.com.                218     IN      TXT      "v=spf1 include:_spf.google.com i
nclude:email.chargebee.com include:7168674.spf05.hubspotemail.net ~all"
tryhackme.com.                218     IN      TXT      "google-site-verification=umR4x8H
uzWMF5g3656JY1b-61NuryD0-GqGnYN130No"

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Feb 06 02:45:11 UTC 2025
;; MSG SIZE rcvd: 238

juan_pablo@Lab4Linux:~$ |
```

I tried an nslookup for the three ip addresses provided from the dig command, so I just used nslookup on 8.8.8.8, which resolved to google's DNS server.

```
juan_pablo@Lab4Linux:~$ nslookup 104.22.54.228
** server can't find 228.54.22.104.in-addr.arpa: NXDOMAIN

juan_pablo@Lab4Linux:~$ nslookup 104.22.55.228
** server can't find 228.55.22.104.in-addr.arpa: NXDOMAIN

juan_pablo@Lab4Linux:~$ nslookup 172.67.27.10
** server can't find 10.27.67.172.in-addr.arpa: NXDOMAIN
```

```
juan_pablo@Lab4Linux:~$ nslookup 8.8.8.8
8.8.8.8.in-addr.arpa      name = dns.google.

Authoritative answers can be found from:

juan_pablo@Lab4Linux:~$ |
```

22. Create two VMs (one Windows, the other Linux) and put them in the same VNet (VPC). Show that one VM can ping the other VM, using only the private IP addresses. Document the steps required to accomplish this setup and show the results

Create a virtual machine ...

 [Help me create a low cost VM](#) [Help me create a VM optimized for high availability](#) [Help me choose the right VM size f](#)

[Basics](#) [Disks](#) **[Networking](#)** [Management](#) [Monitoring](#) [Advanced](#) [Tags](#) [Review + create](#)

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution. [Learn more](#)

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network * ⓘ

lab1-vnet

[Create new](#)

Subnet * ⓘ

default (10.0.0.0/24)

[Manage subnet configuration](#)

When creating the virtual machine, it's important to put it in the same virtual network as the other VM, shown above.

Virtual machines ...

Cincinnati State Technical and Community College (livecincinnatistate.onmicrosoft.com)

[+ Create](#) [Switch to classic](#) [Reservations](#) [Manage view](#) [Refresh](#) [Export to CSV](#) [Open query](#) [Assign tags](#) [Start](#)

[Subscription equals all](#) [Type equals all](#) [Resource group equals all](#) [Location equals all](#) [Add filter](#)

Showing 1 to 2 of 2 records.

☐	Name ↑↓	Subscription ↑↓	Resource group ↑↓	Location ↑↓	Status ↑↓	Operating system
☐	 Lab4Linux	Azure for Students	Labs	East US	Running	Linux
☐	 WindowsLab4	Azure for Students	Labs	East US	Running	Windows

Here are both machines running.

 **Lab4Linux** [✕](#) [☆](#) [...](#)

Virtual machine

[Overview](#) [Activity log](#) [Access control \(IAM\)](#) [Tags](#) [Diagnose and solve problems](#) [Connect](#) [Networking](#) [Settings](#)

[Help me copy this VM in any region](#)

[Connect](#) [Start](#) [Restart](#) [Stop](#) [Hibernate](#) [Capture](#) [Delete](#) [Refresh](#) [Open](#)

Virtual machine

Computer name

Lab4Linux

Operating system

Linux (ubuntu 24.04)

VM generation

V2

VM architecture

x64

Networking

Public IP address

172.190.50.245 { Netw
inter

Public IP address (IPv6)

-

Private IP address

10.0.0.4

Private IP address

-

WindowsLab4
Virtual machine

Help me copy this VM in any region

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Connect
- Networking
- Settings
- Availability + scale
- Security

Connect
Start
Restart
Stop
Hibernate
Capture
Delete
Refresh

time created

2/6/2025, 3:03 AM UTC

Tags

[\(edit\)](#)
[Add tags](#)

Properties
Monitoring
Capabilities (8)
Recommendations
Tutorials

Virtual machine

Computer name	WindowsLab4
Operating system	Windows (Windows Server 2019 Datacenter)
VM generation	V2

Networking

Public IP address	20.115.60.122 (Network interface v
Public IP address (IPv6) -	
Private IP address	10.0.0.5

The Linux machine has a local ip of 10.0.0.4, and the Windows machine has an ip of 10.0.0.5.

On the Windows machine, the ICMP Rule needs to be turned on, as it is currently off:

20.115.60.122 - Remote Desktop Connection

Windows Defender Firewall with Advanced Security

File
Action
View
Help

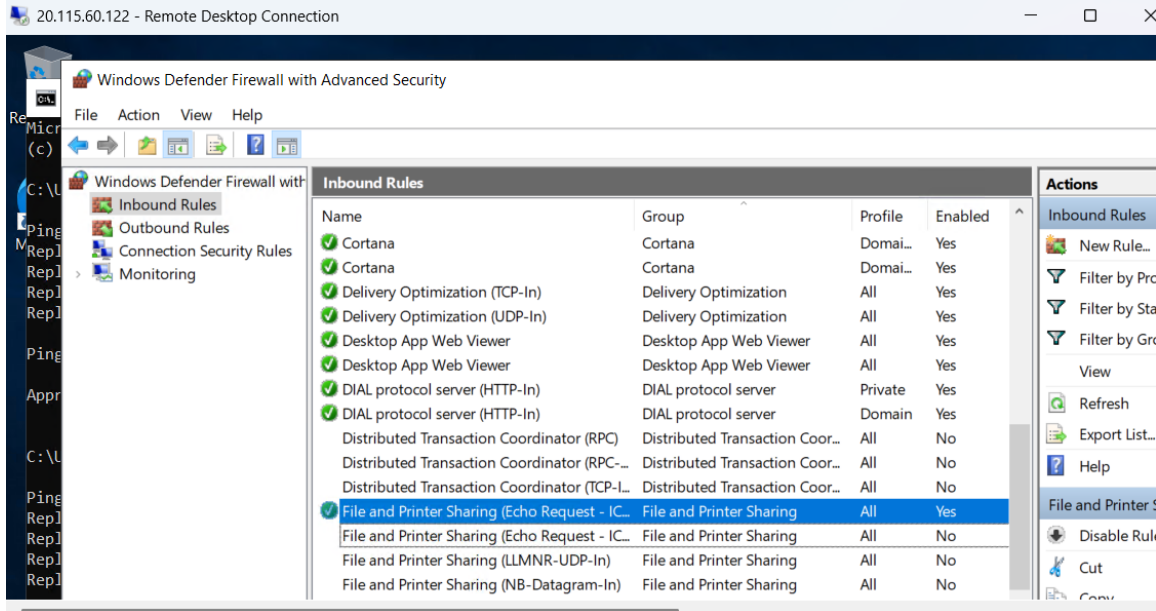
Inbound Rules

Outbound Rules

Connection Security Rules

Monitoring

Name	Group	Profile	Enabled
Cortana	Cortana	Domai...	Yes
Cortana	Cortana	Domai...	Yes
Delivery Optimization (TCP-In)	Delivery Optimization	All	Yes
Delivery Optimization (UDP-In)	Delivery Optimization	All	Yes
Desktop App Web Viewer	Desktop App Web Viewer	All	Yes
Desktop App Web Viewer	Desktop App Web Viewer	All	Yes
DIAL protocol server (HTTP-In)	DIAL protocol server	Private	Yes
DIAL protocol server (HTTP-In)	DIAL protocol server	Domain	Yes
Distributed Transaction Coordinator (RPC)	Distributed Transaction Coord...	All	No
Distributed Transaction Coordinator (RPC-...	Distributed Transaction Coord...	All	No
Distributed Transaction Coordinator (TCP-I...	Distributed Transaction Coord...	All	No
File and Printer Sharing (Echo Request - IC...	File and Printer Sharing	All	No
File and Printer Sharing (Echo Request - IC...	File and Printer Sharing	All	No
File and Printer Sharing (LLMNR-UDP-In)	File and Printer Sharing	All	No



After enabling the rule, above, you get ping replies on the Ubuntu machine:

```
windows server: 10.0.0.5
ubuntu: 10.0.0.4
```

```
20.115.60.122 - Remote Desktop Connection
```

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.17763.6775]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\juan_pablo>ping 10.0.0.4

Pinging 10.0.0.4 with 32 bytes of data:
Reply from 10.0.0.4: bytes=32 time=1ms TTL=64
Reply from 10.0.0.4: bytes=32 time=1ms TTL=64
Reply from 10.0.0.4: bytes=32 time=1ms TTL=64
Reply from 10.0.0.4: bytes=32 time=1ms TTL=64

Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\juan_pablo>ping 10.0.0.4

Pinging 10.0.0.4 with 32 bytes of data:
Reply from 10.0.0.4: bytes=32 time=1ms TTL=64
Reply from 10.0.0.4: bytes=32 time=1ms TTL=64
Reply from 10.0.0.4: bytes=32 time=1ms TTL=64
Reply from 10.0.0.4: bytes=32 time=1ms TTL=64
```

```
azureuser@Lab4Linux:~$ curl -sL https://aka.ms/InstallAzureCLIDeb | sudo bash
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
```

The Windows server can also ping the Ubuntu machine, showed above on the left.

23. Show that you can copy files to/from the BLOB file share from your Linux VM

```
azureuser@Lab4Linux:~$ curl -sL https://aka.ms/InstallAzureCLIDeb | sudo bash
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
```

First I downloaded and installed the azure CLI module.

```
azureuser@Lab4Linux:~$ az login
To sign in, use a web browser to open the page https://microsoft.com/devicelogin
and enter the code A4KCH2KW4 to authenticate.

Retrieving tenants and subscriptions for the selection...

[Tenant and subscription selection]

No      Subscription name      Subscription ID      Tenant
-----
-----
[1] *   Azure for Students  15000e5d-c82a-486f-931f-e8d8f2f20a32  Cincinnati State Technical and Community College
```

Then I logged in with my account.

```
azureuser@Lab4Linux:~$ az storage blob upload --account-name lab3blobjpdL --container-name lab4 --file PublicFiles/file.txt --name file.txt --connection-string "DefaultEndpointsProtocol=https;EndpointSuffix=core.windows.net;AccountName=lab3blobjpdL;AccountKey=psF17q2UHiKFe0u4ApFc3MF0jbc+Cdi0j9mU5Sc2bEq25yHBY3g+9NWxwziNadz8X9G+D9sZBXYL+AStxB6Wrw==;BlobEndpoint=https://lab3blobjpdL.blob.core.windows.net/;FileEndpoint=https://lab3blobjpdL.file.core.windows.net/;QueueEndpoint=https://lab3blobjpdL.queue.core.windows.net/;TableEndpoint=https://lab3blobjpdL.table.core.windows.net/"
```

Next I put in the command to upload the file file.txt into the blob.

lab4

Container

Search

Upload

Change access level

Refresh

Delete

Change tier

Acquire lease

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: lab4

Search blobs by prefix (case-sensitive)

Add filter

Name	Modified	Access tier
☐ de Legarreta Lab 3 Part 2 2025.docx	2/5/2025, 10:51:36 PM	Hot (Inferred)
☐ file.txt	2/5/2025, 11:14:43 PM	Hot (Inferred)

Here is the file in the blob!

```

azureuser@Lab4Linux:~$ az storage blob download --account-name lab3blobjpd1 --con
tainer-name lab4 --name file.txt --sas-token sp=r&st=2025-02-06T04:28:45Z&se=2025
-02-06T12:28:45Z&spr=https&sv=2022-11-02&sr=b&sig=lpaxKx%2FWIPlsboigQwjZg037Gw72j
TREWCD8Lj5CAkk%3D
[1] 7302
[2] 7303
[3] 7304
[4] 7305
[5] 7306
[6] 7307
[2] Done st=2025-02-06T04:28:45Z
[3] Done se=2025-02-06T12:28:45Z
[4] Done spr=https
[5] Done sv=2022-11-02
azureuser@Lab4Linux:~$ ls
PublicFiles
[1]- Done az storage blob download --account-name lab3blobjpd
l --container-name lab4 --name file.txt --sas-token sp=r
[6]+ Done sr=b
azureuser@Lab4Linux:~$

```

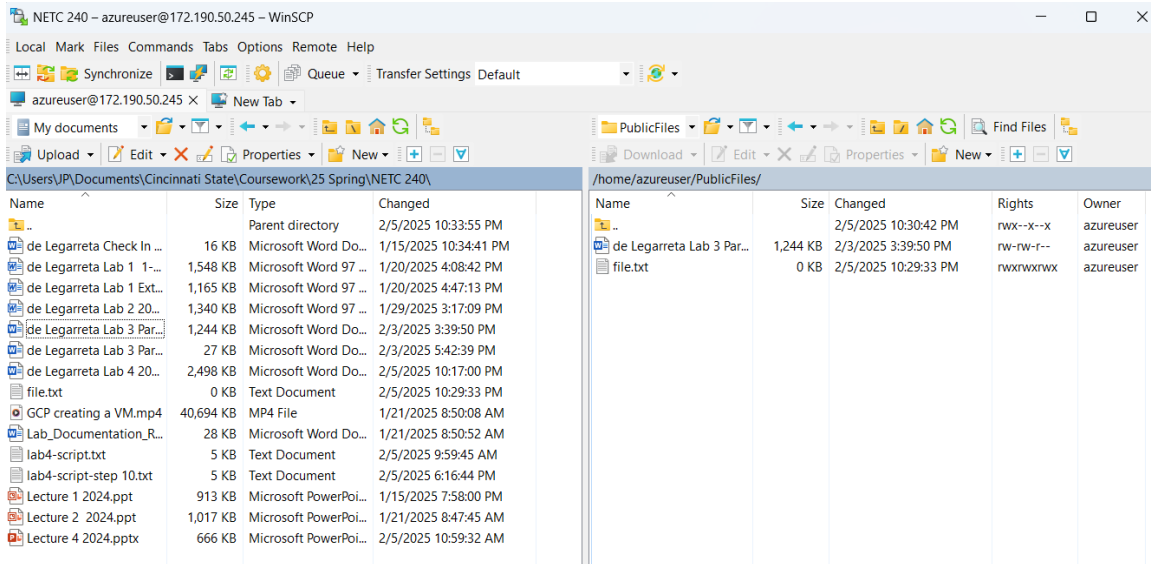
Here's the command to retrieve that same file, and an ls to show the file in the directory!

24. Use an FTP client to create a SFTP connection to your VM. Copy the files from your Azureuser account to your PC. Then copy a file from your PC to the Azureuser home directory.

Name	Size	Type	Changed
..		Parent directory	2/5/2025 5:05:30 PM
known_hosts	2 KB	File	2/5/2025 5:05:30 PM
known_hosts.old	1 KB	OLD File	2/5/2025 5:05:27 PM
Lab3Linux_key.pem	3 KB	PEM File	1/29/2025 8:38:45 PM
Lab3Linuxpriv.ppk	3 KB	PuTTY Private Key ...	1/29/2025 11:05:50 PM
Lab3Linuxpub	1 KB	File	1/29/2025 11:05:30 PM
Lab3Linuxpub.ppk	1 KB	PuTTY Private Key ...	1/29/2025 11:07:30 PM

Name	Size	Changed	Rights	Owner
file.txt	0 KB	2/5/2025 10:29:33 PM	rw-rw-rw-	azureuser

Successfully accessed the VM through WinSCP over SFTP.



I copied a file called file.txt from the VM to my local machine, and I copied one of the Lab 3 files from my computer to the VM.

Turn in:

MS Word, which includes annotated screenshots and explanations that shows the “proof of work” for each step. Label each step 1,2,.. Include the answers to the Questions at the end of the document.